

Start-In-Motion (SIM) Guide

Power Application Controller™





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1 INTRODUCTION

Start-In-Motion (SIM) is a feature used to enable the FOC firmware to start a motor while the motor is in motion. Normally the Align&Go method of starting up requires that the motor be aligned to a known position from which motion will then start. But if the motor align is started while the motor is in motion, then the alignment and consequently the startup may fail. SIM handles this case by detecting if the motor is in motion and then modifies the startup procedure to ensure that the startup is successful.

This feature is specifically useful in the following cases:

- 1) Applications where the motor may be disabled and quickly re-enabled before the motor has come to a full stop.
- 2) Applications where an external force may be causing the motor to already be free-spinning at the time that it is enabled.
- 3) Applications where the motor may need to be quickly restarted in the reverse direction.

There are 3 basic methods for implementing SIM:

Voltage-Mode

- 1) Samples phase voltage to determine if the motor is in motion.
- 2) Requires 3 ADC channels for measuring phase voltages.
- 3) Voltage- mode stop detection requires a maximum low-side driver ON time during the SIM control period. As a result, the maximum braking force is not achievable.
- 4) Available for 3-shunt and 1-shunt configuration.
- 5) Available for sensorless motor position estimator algorithms.
- 6) The voltage vector magnitude during an in-motion start (SIM2CL) can be determined using either the phase voltage measurement (BEMF) or the motor's K_e parameter together with the speed.

Current-Mode

- 1) Samples phase currents to determine if the motor is in motion.
- 2) Eliminates the need for sampling 3 phase voltages resulting in simpler PCB since it uses the current sampling channels already used for motor control.
- 3) Does not have a maximum low-side driver ON time limitation and therefore maximum braking force is achievable.
- 4) Requires additional tuning.
- 5) **Currently it is available only for 3-shunt configuration.**
- 6) Available for sensorless motor position estimator algorithms.
- 7) The voltage vector magnitude during an in-motion start (SIM2CL) can only be determined using the motor's K_e parameter together with the speed.

Hall-Mode

- 1) Uses the existing hall engine in hall-sensor based motor control algorithms.
- 2) Does not require use of any additional ADC channels.
- 3) Does not have a maximum low-side driver ON time limitation and therefore maximum braking force is achievable.
- 4) Available for 3-shunt and 1-shunt configuration.
- 5) Available only for hall-sensor based motor control algorithms.
- 6) The voltage vector magnitude during an in-motion start (SIM2CL) can only be determined using the motor's K_e parameter together with the speed.

2 DESCRIPTION

SIM detection mode is entered as soon as the motor is enabled. If the motor is determined to be in motion, it is also constantly determining the direction of motion as well as calculating the rotor speed. When the motor is enabled, the SIM values are then used to determine the steps to take for startup.

There are main two main aspects to SIM detection:

- 1) Initial Motion Detection
When a motor is first enabled, it needs to detect if the motor is already in motion or not before the system continues the startup sequence. This includes determining rotor direction and speed.
- 2) Braking Stop Detection
When SIM determines that it needs to brake, while it is braking it needs to be able to detect that the motor has successfully stopped before continuing with the startup sequence.

3 FIRMWARE FLOW

When SIM mode is entered, the sequencers are reconfigured for either for phase voltage sampling or for phase current sampling, or the hall interrupts are configured, depending on the method chosen. The sequencers are then reconfigured for motor control operation when the motor is enabled.

When the controller is in SIM mode, it is constantly observing the motor to determine the status. From the flowchart in Figure 1 it can be concluded that the possible motor status outcomes that SIM can return are as follows:

STATUS_SIM_MOTIONNOTDETECTED

The BEMF voltage generated by the motor is too low to be properly detected by the ADC. This condition is interpreted as the motor not moving. In this case the startup sequence can proceed as normal.

STATUS_SIM_SLOWSPEED

The motor is determined to be in the same direction of motion as the direction command, but the speed is below the OpenLoop-Closed Loop switchover speed.

STATUS_SIM_WRONGDIRECTION

The motor is determined to be spinning in a direction opposite to that of the direction command.

STATUS_SIM_SWITCH2CL

The motor is determined to be spinning in the same direction as the direction command, and the speed is above the OpenLoop-Closed Loop switchover speed. In this case SIM instructs FOC to jump in and start controlling the motor immediately skipping the open-loop sequence.

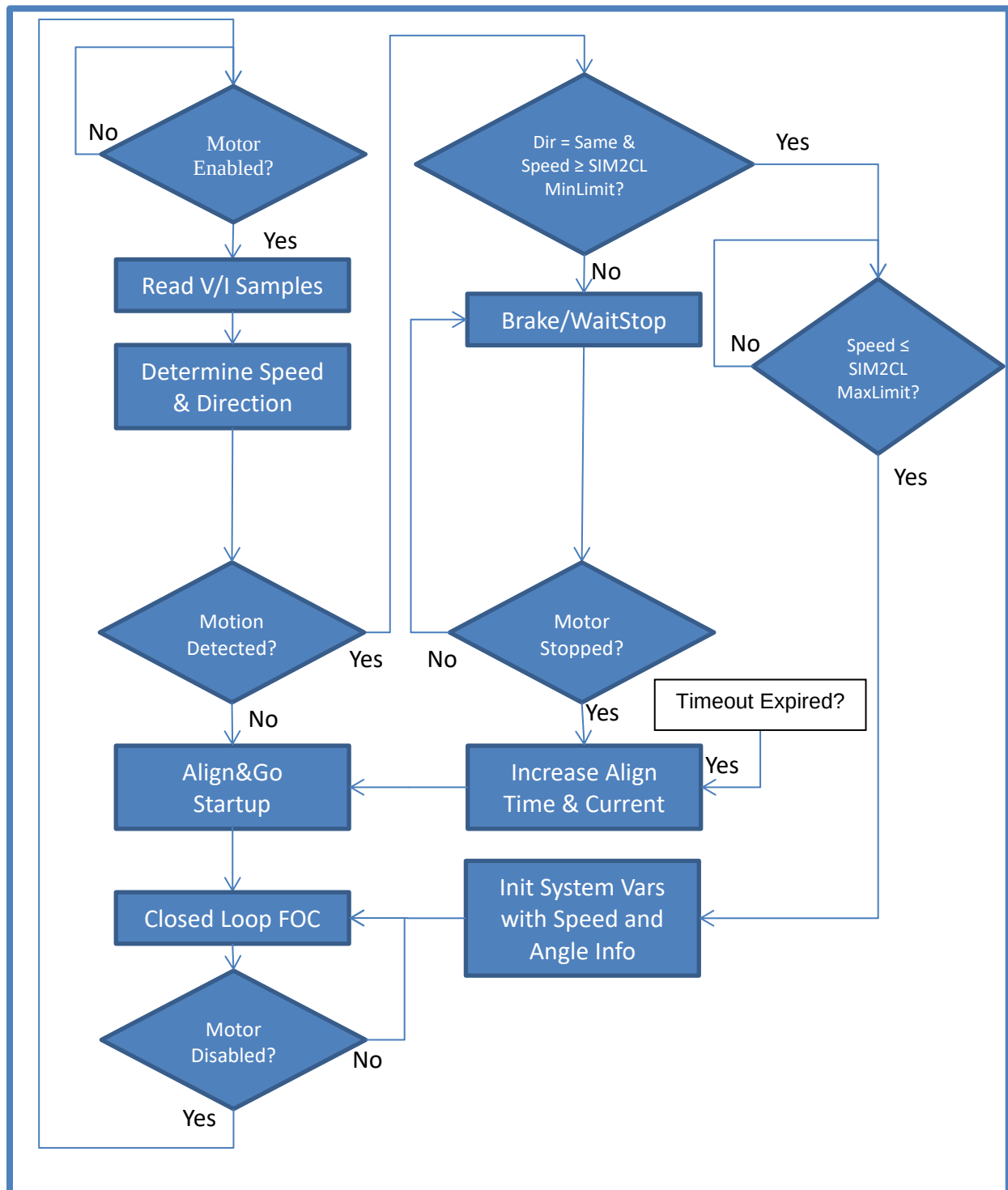


Figure 1: SIM Flowchart



When the user enables the motor, then SIM makes a decision on how best to proceed with the startup based on the status of the motor at that moment.

The user has the ability to configure the responses to some of the SIM cases listed above as shown below:

- 1) The following SIM cases can be independently defined as either “Brake” events or “WaitStop” events:
 - a. **STATUS_SIM_SLOWSPEED**
This case can be defined either as a “Brake” event in which SIM will actively brake the motor, or as a “WaitStop” event in which SIM will just wait until the motor slows down on its own until motion has stopped. Configuration of this SIM condition as a “Brake” event is done by the ENABLE_SIMBRAKE_SLOWSPEED compile option.
 - b. **STATUS_SIM_WRONGDIRECTION**
This case can be defined either as a “Brake” event in which SIM will actively brake the motor, or as a “WaitStop” event in which SIM will just wait until the motor slows down on its own until motion has stopped. Configuration of this SIM condition as a “Brake” event is done by the ENABLE_SIMBRAKE_REVDIRECTION compile option.
 - c. **STATE_MAIN_BRAKE**
This is not officially a SIM case but it is a case that can be selected by the main state machine when the motor is disabled. The user can select to define the disabling of the motor as a “Brake” event, so that the braking mechanism in SIM can be activated every time that the user disables the motor. This option is selected by the ENABLE_SIMBRAKE_MOTORDISABLE compile option.

The SIM state machine controls the firmware flow depending on how the user configures the SIM responses. This details of this state machine are illustrated in Figure 2 and Figure 3.

Please refer to the Firmware Options section for more details on how to configure these options in the firmware.

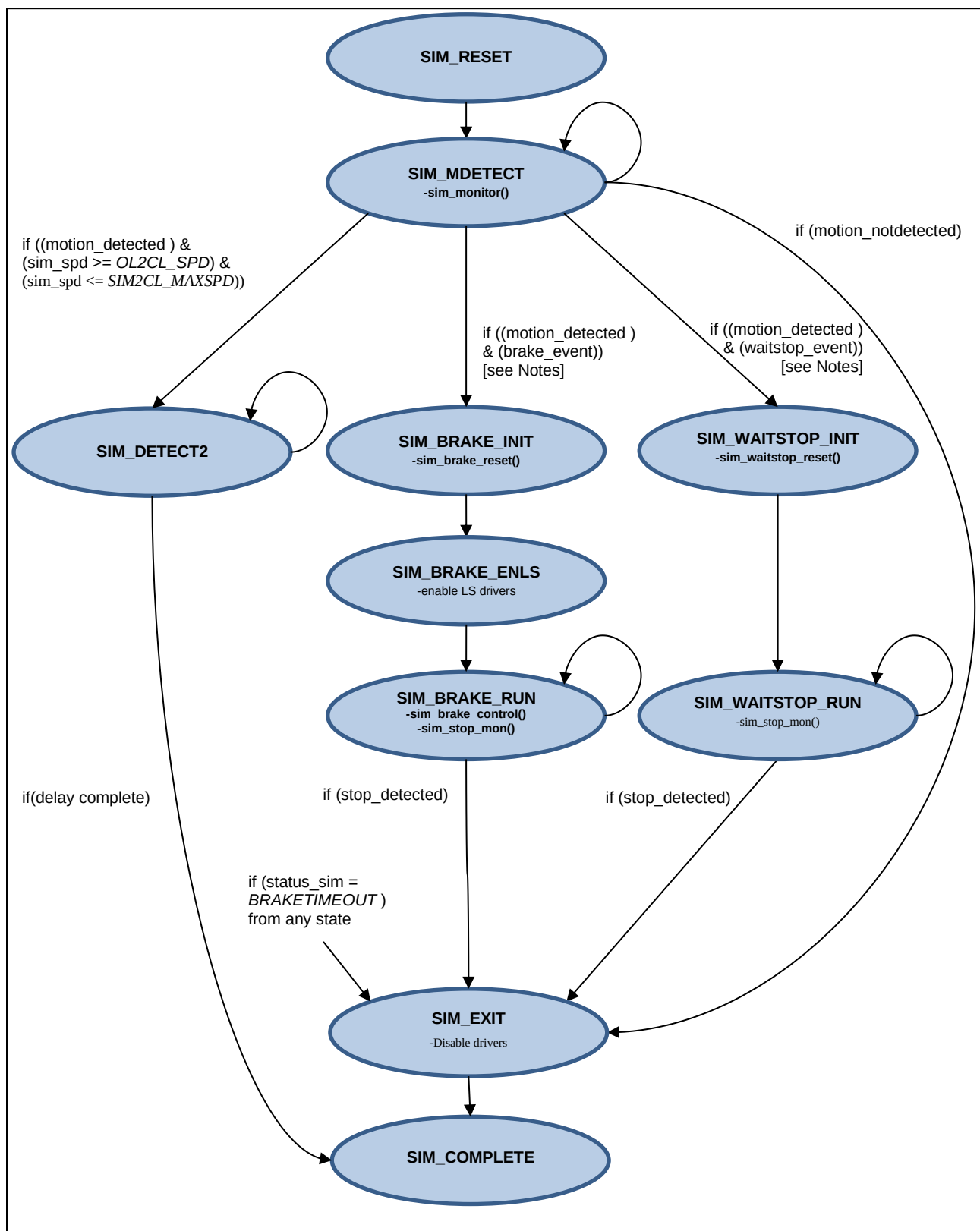


Figure 2: SIM State Machine Diagram (Voltage-Mode & Hall_Mode)

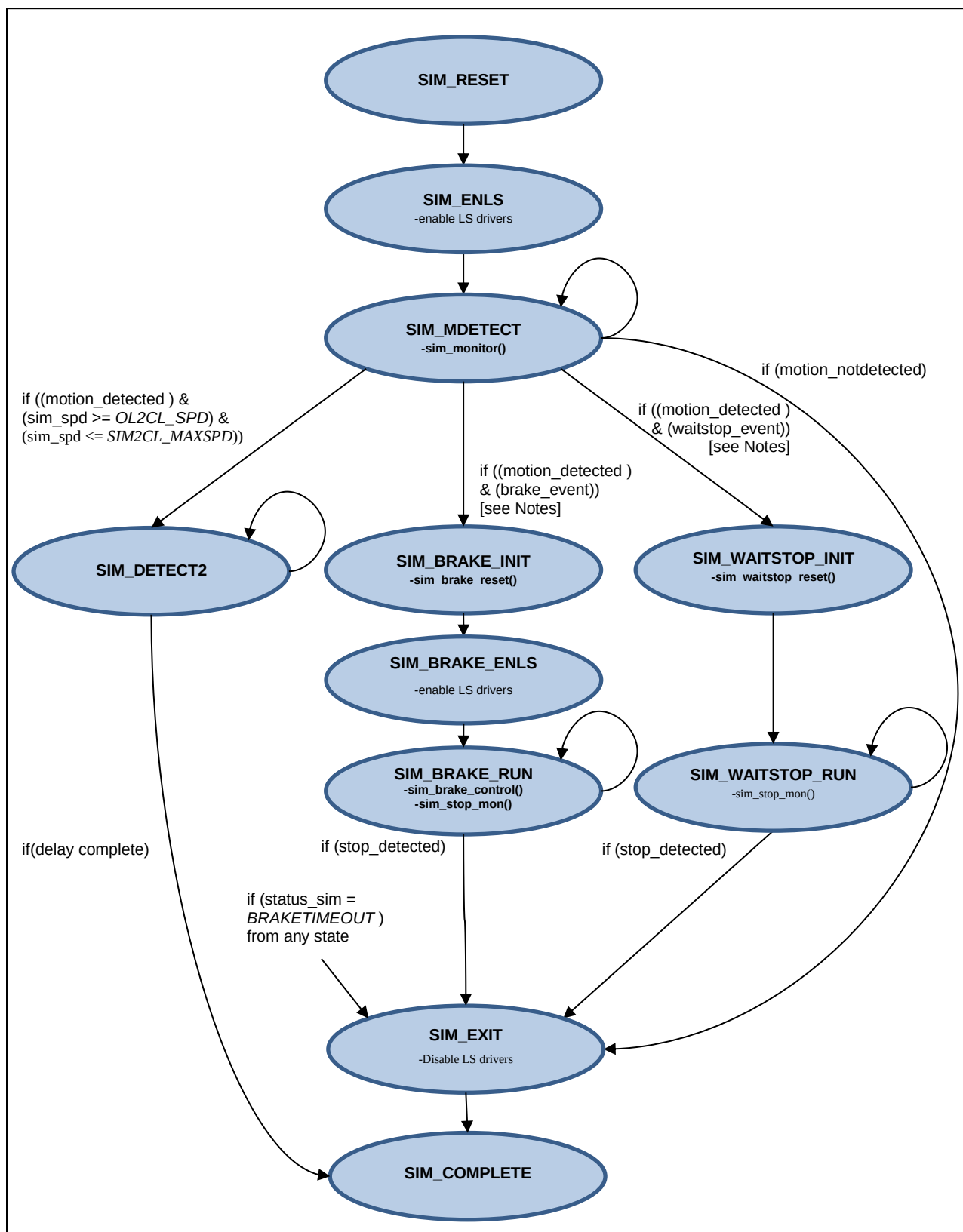


Figure 3: SIM State Machine Diagram (Current-Mode)

4 HARDWARE CONFIGURATION

Voltage-Mode

Voltage- mode SIM requires that the 3 motor phase voltages be sensed for BEMF voltage. Because the phase voltages are above the rated voltage range of the IO pins, this phase voltage needs to be divided down as shown in the schematic capture below.

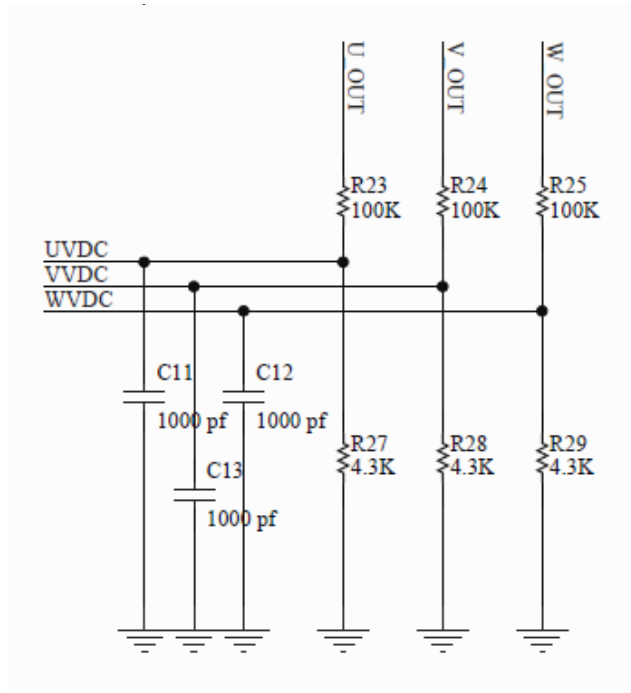


Figure 4: Phase Voltage Sensing Hardware

The ADC voltage input range is 0-2.5V, therefore the divider ratio needs to be chosen properly so that when the phase voltage is at its maximum value (typically Vbus), the ADC voltage is not exceeded.

$$V_{aio} = V_{bus} * \text{Ratio}$$

$$\text{Ratio} = R_{bottom} / (R_{top} + R_{bottom})$$

If the ratio is too small, then the phase voltage may be excessively attenuated making it more difficult to detect low level BEMF voltages and therefore reducing SIM detection at low speeds.

A small capacitor is also added for filtering purposes.

Current-Mode

Current-mode SIM requires no additional hardware. It uses same current sensing shunts as the motor control algorithm.

Hall-Mode

Hall-mode SIM requires no additional hardware. It uses same hall engine as the motor control algorithm.



5 SOFTWARE CONFIGURATION

5.1 AIO Port Configuration

Voltage-Mode

Voltage- mode SIM requires 3 analog IO ports to be reserved for phase voltage sensing. It is recommended that AIO7/8/9 be used for this purpose. To configure these ports for voltage-sensing, they need to be set to amplifier mode with a gain that is appropriate to use with the divider ratio of the sensing voltage divider so that the divided voltage does not exceed the input range of the ADC. Typically this gain is set to a value of 1.

Current-Mode

No additional port configuration is required for current-mode SIM because this method uses the same configuration used by the motor drive FOC algorithm to sense phase current.

Hall-Mode

No additional port configuration is required for Hall-mode SIM because this method uses the same configuration used by the motor drive FOC algorithm to read Hall sensor position.

5.2 Sequencer Configuration

The sequencers need to be configured for SIM whenever SIM mode is entered.

Voltage-Mode

The sequencer is configured to sample the 3 phase voltages as well as the VDC bus voltage. This is currently done by using 2 sequencers, one for the phase voltages and one for the bus voltage.

The details of the sequencer configuration are explained below:

- 1) Sequencer slot assignments: :
 - a. Seq0/N0 : Motor Phase Voltage U
 - b. Seq0/N1 : Motor Phase Voltage V
 - c. Seq0/N2 : Motor Phase Voltage W
 - d. Seq1/N0 : Vbus
- 2) Trigger Mode: S1 follows S0 in chain trigger mode.

S1 sampling the bus voltage is automatically triggered after S0 completes sampling the phase voltages. Therefore only one CCR trigger is necessary for S0.

- 3) ADC ISR Trigger: Sequencer S0

The ADC ISR will start running the SIM control loop as soon as the new phase voltages are sampled and ready to use. It will not wait for the bus voltage to be sampled. This configuration guarantees that any additional sampling the user may need to add to S1 will not delay the start of the control loop.

- 4) Sequencer S0 Trigger CCR: A0
- 5) Sequencer S0 Trigger Edge: Low-to-High
- 6) Sequencer S0 Trigger Value: Minimum allowed voltage-SIM duty cycle

Although the trigger point may not matter during motion detection in voltage- mode SIM, if braking is activated, then voltage can only be sampled during the OFF time of the Low-Side gate drivers. All 3 phase voltage ADC samples must be completed before the Low-Side gate driver transitions to the ON state. Therefore the minimum SIM braking duty cycle is determined by the smallest pulse-width in OFF state that allows the 3 phase voltages to be sampled. The sampling trigger of the phase voltages is set to a fixed value at the beginning of the minimum OFF time. Please refer to Figure 5 for more detail.

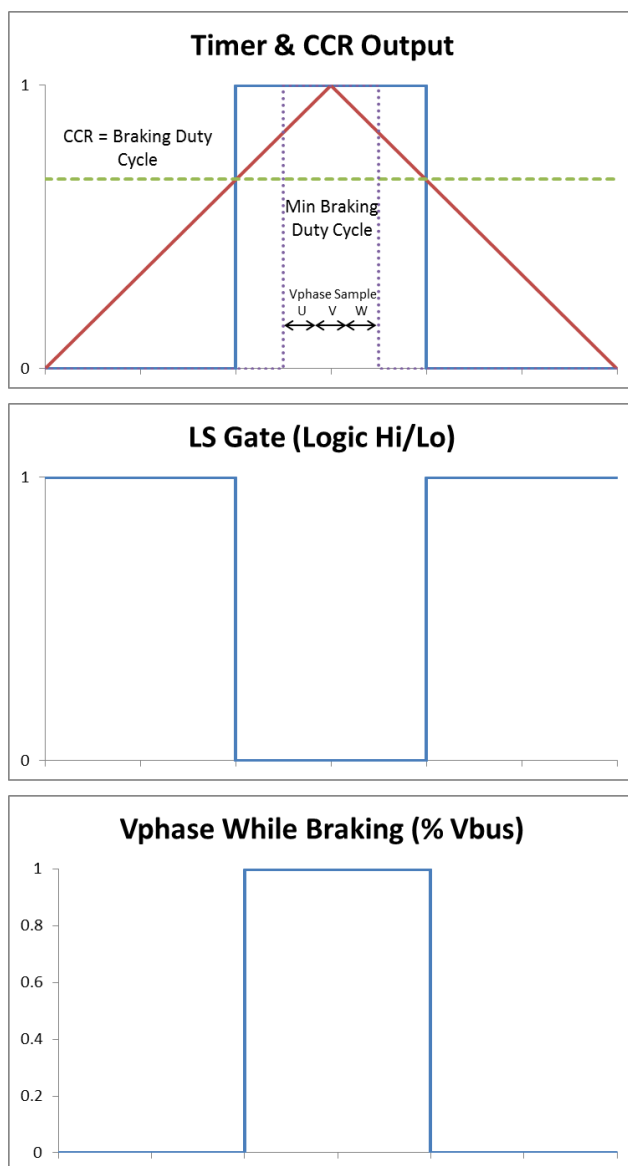


Figure 5: Sampling point in Voltage-Mode SIM while braking

Current-Mode

The sequencer is configured to sample the 3 phase currents as well as the VDC bus voltage. This is currently done by using 2 sequencers, one for the phase currents and one for the bus voltage.

The details of the sequencer configuration are explained below:

- 1) Sequencer slot assignments: :
 - a. Seq0/N0 : Motor Phase Current U (with S/H)
 - b. Seq0/N1 : Motor Phase Current V (with S/H)
 - c. Seq0/N2 : Motor Phase Current W (with S/H)
 - d. Seq1/N0 : Vbus

- 2) Trigger Mode: S1 follows S0 in chain trigger mode.

S1 sampling the bus voltage is automatically triggered after S0 completes sampling the phase currents. Therefore only one CCR trigger is necessary for S0.

- 3) ADC ISR Trigger: Sequencer S0

The ADC ISR will start running the SIM control loop as soon as the new phase currents are sampled and ready to use. It will not wait for the bus voltage to be sampled. This configuration guarantees that any additional sampling the user may need to add to S1 will not delay the start of the control loop.

- 4) Sequencer S0 Trigger CCR: A0

- 5) Sequencer S0 Trigger Edge: Low-to-High

- 6) Sequencer S0 Trigger Value: 4 counts below timer peak (essentially PWM center)

The trigger point is set to the middle of the Low-Side HIGH state. The phase current can only be measured across the shunt when the low-side gate driver signal is in the HIGH state. This maximizes the motion detection sensitivity.

Hall-Mode

Hall-mode does not require any sequencer configuration since it uses the same hall interrupt engine used for motor control.

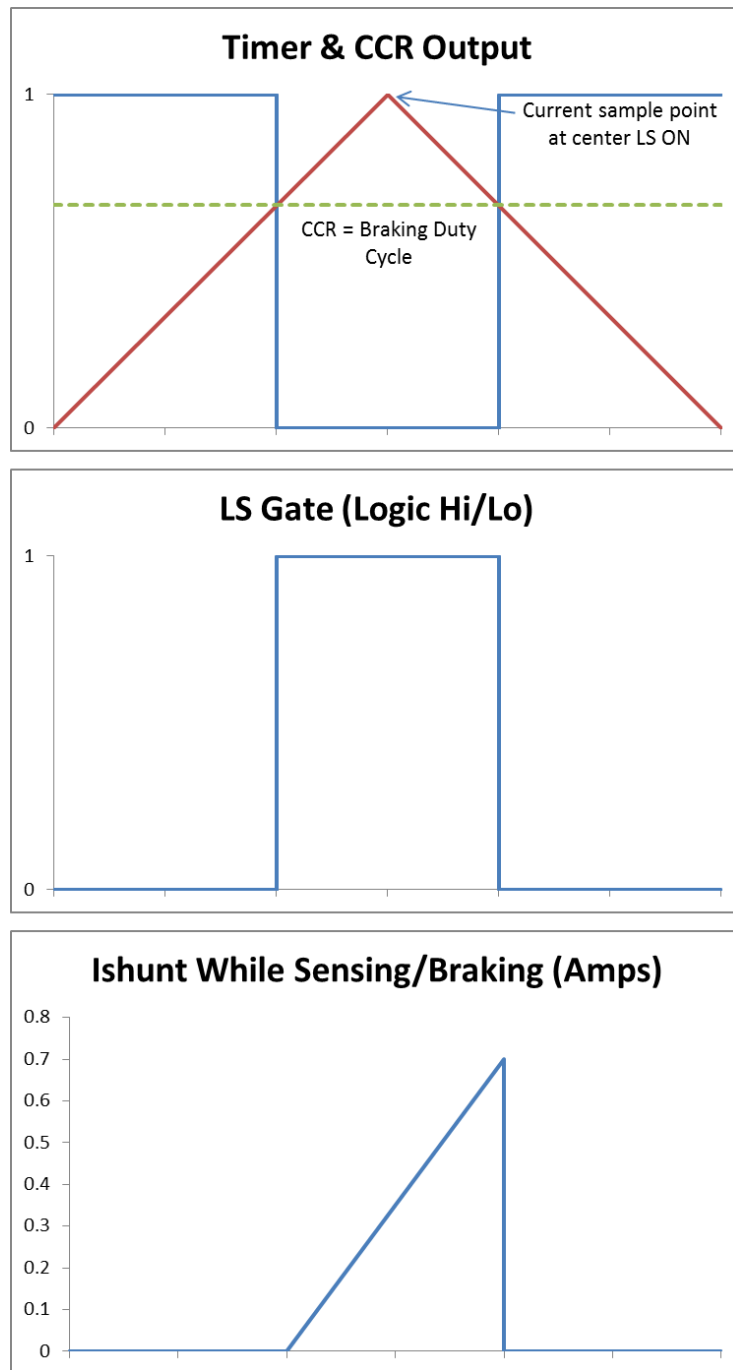


Figure 6: Sampling Point in Current-Mode SIM

6 USING SIM

6.1 Enabling SIM

To enable the SIM feature, find the `ENABLE_SIM` option in `config_features.h` and verify that it is defined.

6.2 Firmware Options

To configure SIM, please verify that the desired options are defined in `config_features.h` and that the selections and values are appropriate for the application:

SIM Method Selection

`SEL_SIM_METHOD`

Use to select SIM method. When running sensorless control algorithms, the available options are:

`SIM_METHOD_VOLTAGE_BEMF`

`SIM_METHOD_VOLTAGE_KE`

`SIM_METHOD_CURRENT`

When using voltage mode SIM, it is preferable to use `Ke` for determining the voltage magnitude because it provides the most reliable results.

When using Hall sensor control, `SIM_METHOD_HALL` is automatically selected as the only option.

SIM Operation Modes

`ENABLE_SIMBRAKE_MOTORDISABLE`

SIM will activate braking every time that the motor is disabled. If this feature is not enabled, motor will not brake, just free spin to a stop. When braking at high speeds, braking current can become very high. To keep this current under control, it is recommended to slow down the braking ramp rate by increasing the `BRAKING_DUTY_INC_SKIP_COUNT` value.

`ENABLE_SIMBRAKE_REVDIRECTION`

SIM will activate braking every time that reverse direction is detected when starting the motor. If this feature is not enabled, motor will not brake, just free spin and wait until motor stops before it starts. When braking at high speeds, braking current can become very high. To keep this current under control, it is recommended to slow down the braking ramp rate by increasing the `BRAKING_DUTY_INC_SKIP_COUNT` value.

`ENABLE_SIMBRAKE_SLOWSPEED`

SIM will activate braking every time that slow speed (below switchover speed) is detected when starting the motor. If this feature is not enabled, motor will not brake, just free spin and wait until motor stops before it starts.

SIM Configuration

ENABLE_SIMANGLEBLEND

When SIM2CL is called, the SIM angle is gradually blended with the estimated angle during a transition period for a smoother transition. This is automatically enabled when running sensorless control algorithms to help smoothen out the transition.

6.3 Firmware Parameters

SIM detection and braking uses the firmware specified configuration parameters below. These parameters are defined in `sim_funcs.h`:

SIM_CONTROL_DIVIDER

This is the control loop frequency divider relative to the PWM frequency when running SIM. The default value is “2”, which means that if the PWM frequency is selected to be 20kHz the SIM control frequency is 10kHz. This value may need to change depending on the PWM frequency and the device family used. For 52xx family devices it is recommended that the SIM control loop period is at least 100us. For 55xx family devices it is recommended that the SIM control loop period is at least 50us. If the GUI reports a warning indicating that the control loop time is exceeded while doing a SIM2CL transition it is a sign that the divider value needs to be increased in order to reduce the SIM control loop frequency.

SIM_ANGLE_BLEND_MAX_CNT

Number of cycles to blend angle during SIM2CL transition.

SIM BRAKING_DUTYRAMP_TIME_SEC_Q8

Time to ramp braking duty cycle to maximum braking force (min/max = 0.1/5 sec)
This is the length of time during which the braking duty cycle ramp will complete.

SIM_IQREF_MIN

Minimum iq reference allowed during SIM2CL to prevent any negative torque due to measurement inaccuracy from charging Vbus. Value is a percentage of total current scale in decimal format.

SIM_SIM2CL_MAXSPD_PCT_Q16

Maximum speed as percentage of maximum CL speed that SIM2CL is allowed to trigger. If speed is higher than this limit SIM will wait until speed is below this threshold before triggering SIM2CL.

SIM_SIM2CL_PWM_XTRACOUNTS

This is the number of FOC control cycles that will be compensated during a SIM2CL transition. This is necessary to compensate for internal delays in the handling of the duty cycle values.

SIM_MDETECT_DETECTTIME_SEC_Q16 (Voltage and Current SIM only)

Length of time during which motion will be detected.

SIM_DETECT2_SPEEDFILTER_FC_HZ_Q16 (Voltage and Current SIM only)

Cutoff frequency for speed filter during SIM2CL transition. Increasing the filter cutoff frequency increases the detection time.

SIM_MDETECT_HALLEDGENUM *(Hall SIM only)*

Number of hall transition edges to detect before declaring that motion is detected. Motion is declared to be not detected when the hall timer expires. Hall timer is configured in config_app.h.

SIM_HALLFILTER_SHIFT *(Hall SIM only)*

Hall speed filter shift depth value. Increasing the value will make the filter more aggressive. The range of values are 0-5 where “0” value means no filtering is applied. The filter applied is described below:

$$y[n+1] = y[n] + (x[n] - y[n])/2^{\text{SIM_HALLFILTER_SHIFT}}$$

SIM_MDETECT_DUTYRAMP_TIME_SEC_Q8 *(Current SIM only)*

Time to ramp motion detection duty cycle to final value before declaring no motion.

These parameters are defined in the respective board *.h files:

SIM_BEMF_DET_THRESHOLD_ADCCNT_V**SIM_BEMF_DET_THRESHOLD_ADCCNT_I**

Threshold for minimum phase-to-CT voltage or current (depending on which mode is chosen) in ADC counts. Phase voltages measuring below this value are treated as if the voltage is 0V. This value needs to be above the noise floor of the phase voltage conversion for SIM detection to work properly. If this value is below the noise floor, SIM will not be able to detect that the motor is stopped. If the value is too high, SIM may incorrectly detect that the motor has stopped before it actually is stopped.

6.4 Tuning Parameters (GUI)

The following parameters are adjustable from the Features->SIM tab in the GUI:

Align Time (s)

Align time (seconds) used every time motor is started after SIM braking had to be activated to stop the motor. It overrides the align time value specified in the Align&Go Tuning section. In the case the motor is still moving after a braking timeout or stop condition was incorrectly determined, a longer align time will help to stop the motor enough for a proper startup.

OL Current Multiplier (x)

Gain multiplier used to increase the total align current every time motor is started after SIM braking had to be activated to stop the motor. The gain is multiplied by the total reference current value entered in the Align&Go Tuning section. In the case the motor is still moving after a braking timeout or stop condition was incorrectly determined, a higher align current will help to stop the motor enough for a proper startup.

Detection Timeout (s)

SIM timeout time (seconds). If SIM does not complete within this time, the SIM process will be aborted and the motor will continue with the startup sequence. In this case, the align time and the align current will be modified as described above. This value needs to be larger than (maybe 2x) the time it normally takes for the motor to slow down to a stop.

V-SIM BEMF Amp Gain (x)

Hardware amplifier gain applied to the phase voltage signal before it is converted by the ADC. For best SIM detection, a gain value of x1 should be assumed and the resistor divider ratio should be chosen accordingly so that a gain of x1 will not exceed 2.5V at the ADC input.

Additionally the following parameter can be adjusted from the Motor Config panel in the GUI specifically for v-SIM using Ke or for i-SIM:

Ke (V*sec/rad)

The BEMF constant is used in conjunction with detected motor speed to determine the correct voltage vector magnitude to apply during a SIM2CL transition. This value should be adjusted is necessary to achieve the smoothest transition possible when configuring SIM in v-SIM mode using Ke or in i-SIM. This parameter is not used when selecting v-SIM using BEMF.



CONTACT INFORMATION

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